

Go: The Simple Programming Language

This is an introduction to Go, an open source programming language that makes it easy for developers to build simple, reliable and efficient software. The latest stable version of Go, the sixth major release so far, is version 1.5, which was released on August 19, 2015.

The Go programming language was developed at Google by Robert Griesemer and UNIX luminaries Rob Pike and Ken Thompson. It is an open source, high-level, compiled, highly efficient, strongly statically typed programming language. Go supports both procedural and object-oriented programming, though its object-orientation is quite different from other object-oriented programming languages like C++, Java, etc.

Some of the features of Go that we will look at are: object-orientation without class, inheritance and overloading, duck typing, closure, concurrency; built-in support for slices, maps and strings; and fast compilation, among others. Let's install Go first. I will describe the installation steps on Ubuntu 15.04, though it can be installed on any distribution of your choice. Go can be installed either from the binary as follows:

```
$ sudo apt-get install golang
```

...or from the Go source code. I will cover the second method of installation, step-by-step.

Step 1: Install the prerequisites package. We need Git to complete the installation.

```
$ sudo apt-get install git
```

Step 2: Download the Go set-up tree. The Go source code is written in Go itself; so to create the bootstrap, we need a workable Go set-up. Download the Go 1.4 set-up tree and configure it as follows:

```
$ wget https://storage.googleapis.com/golang/go1.4.linux-386.tar.gz
$ mkdir $HOME/bootstrap
$ tar -C $HOME/bootstrap -xzf go1.4.linux-386.tar.gz
$ export GOROOT_BOOTSTRAP=$HOME/bootstrap/go*
```

Step 3: Get the Go source and compile it as follows:

```
$ mkdir $HOME/source
$ cd $HOME/source
$ git clone https://go.googlecode.com/go
```



```
$ cd go
$ git checkout go1.5
$ cd src
$ ./all.bash
```

This takes some time. If everything goes right, you should get the message 'ALL TESTS PASSED' along with some other responses. Now, add the Go *bin* directory to your PATH environment variable, as follows:

```
$ export PATH=$PATH:$HOME/source/go/bin*
```

Step 4: Check for the Go installation, by using the following command:

```
$ go version
go version go1.5 linux/386
```

Your actual output may vary, depending on your computer architecture.

Step 5: Create the GOPATH environment variable, as follows:

```
$ mkdir $HOME/workspace
$ export GOPATH=$HOME/workspace*
```

Step 6: Install the Go tool *godoc*, by issuing the following code:

```
$ go get golang.org/x/tools/cmd/godoc
```